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Positive Polynomial Phase

The fixed cubic remains open; a non-special higher-degree family is proved

Offline research checkpoint. No internet search or external retrieval was used. The supplied reports and the frozen results in the research prompt were treated as primary input. In particular, the earlier prime, prime-power, singular-scaling, and residual-critical branches were not re-opened.

The entropy test used throughout is the following. If

$$|A_d| = e^{ad+o(d)}, \qquad |P(A_d, A_d)| \le e^{bd+o(d)},$$

then a fixed power saving is obtained exactly when $b < 2a$, in which case

$$c = 2 - \frac{b}{a} > 0.$$

A. Fixed polynomial conjecture

Let

$$P_3(x, y) = (x - y)^3 - 3(x + y).$$

The Cubic Polynomial Small-Image Conjecture (CPSI) asks whether there are a constant $c_3 > 0$ and arbitrarily large finite $A \subset \mathbb{R}$ such that

$$|P_3(A, A)| \le |A|^{2-c_3}.$$

Verdict for this checkpoint

CPSI is not proved or disproved here. No construction examined in this cycle gives $b < 2a$. The strongest unconditional positive result is instead the following explicit companion theorem.

Main unconditional theorem

Let

$$R(x, y) = 5(x + y) + (x - y)^2,$$

and assume the frozen theorem supplying a constant $c_0 > 0$ and arbitrarily large finite sets $A_n \subset \mathbb{R}$ for which

$$|R(A_n, A_n)| \le |A_n|^{2-c_0}.$$

For every nonconstant $\Phi \in \mathbb{R}[t]$, every $a \in \mathbb{R} \setminus \{0\}$, and every $b \in \mathbb{R}$, define

$$Q_{\Phi,a,b}(x, y) = \Phi\big(R(ax + b, ay + b)\big).$$

Then there are arbitrarily large finite $A'_n \subset \mathbb{R}$ satisfying

$$|Q_{\Phi,a,b}(A'_n, A'_n)| \le |A'_n|^{2-c_0}.$$

Moreover, every such $Q_{\Phi,a,b}$ lies outside both polynomial forms

$$F(g(x) + h(y)), \qquad F(g(x)h(y)),$$

when g and h are required to be nonconstant univariate polynomials. In particular,

$$P_4(x, y) = R(x, y)^2 + R(x, y)$$

is an explicit degree-four, non-special polynomial with a fixed power-saving construction.

B. Non-special verification for the cubic

We first record a differential obstruction that will also handle the companion family.

Lemma B.1: derivative-ratio obstruction

Let $Q \in \mathbb{R}[x, y]$ have nonzero partial derivatives. If

$$Q(x, y) = F(g(x) + h(y))$$

or

$$Q(x, y) = F(g(x)h(y)),$$

where F is nonconstant and both g, h are nonconstant univariate polynomials, then the rational function Q_x/Q_y is multiplicatively separable:

$$\frac{Q_x}{Q_y} = A(x)B(y)$$

for some nonzero rational functions A, B.

Indeed, the two cases give respectively

$$\frac{Q_x}{Q_y} = \frac{g'(x)}{h'(y)}$$

and

$$\frac{Q_x}{Q_y} = \frac{g'(x)h(y)}{g(x)h'(y)}.$$

A nonconstant rational function $\rho(x-y)$ cannot have the form $A(x)B(y)$. To see this, take the logarithmic derivative in x and then differentiate in y. One obtains

$$\left(\frac{\rho'}{\rho}\right)' = 0.$$

Thus ρ'/ρ is constant. At infinity the logarithmic derivative of a nonzero rational function is $O(1/t)$, so that constant is zero and ρ itself is constant, a contradiction.

For $u = x - y$,

$$(P_3)_x = 3(u^2-1), \qquad (P_3)_y = -3(u^2+1),$$

and hence

$$\frac{(P_3)_x}{(P_3)_y} = \frac{1-u^2}{1+u^2}.$$

This is a nonconstant rational function of $x-y$. Lemma B.1 proves that P_3 is in neither special form.

There is also a short degree check. In the additive case, $\deg_x P_3 = \deg_y P_3 = 3$ forces either a linear outer polynomial, which cannot create the mixed cubic terms, or a cubic outer polynomial with linear g, h. The leading form then forces $g+h$ to be proportional to $x-y$, contradicting the term $-3(x+y)$. In the multiplicative case, writing $d = \deg F$, $e = \deg g$, and $f = \deg h$ gives $de = df = 3$, while the total degree would be $d(e+f) = 6$, not 3.

C. Results of the parallel mechanisms

The mechanisms were attacked independently and then audited against the full Cartesian requirement.

- **Algebraic identity, norm, and trace.** The polynomial is a coordinate and has exact Chebyshev, determinant, trace-cube, and varying-norm identities. No fixed nontrivial polynomial field-norm representation exists. Every identity found retains a pair label carrying quadratic entropy.
- **Unit-distance and number fields.** A genuine same-set norm-one trace construction gives an exponentially bounded full image, but its certified output base exceeds the squared input base by more than $128D^2$ per degree.
- **Direct sum-product insertion.** Substitution into the frozen $A = GP$ set gives a natural monomial-coordinate code with exactly $|G|^2|P|^2 = |A|^2$ labels. The archimedean box is worse.
- **High-dimensional digits.** Transcendental digit strings are asymptotically pair-injective, while good reduction modulo 3 gives exact injectivity. These real models have $b = 2a$.
- **Formal tensor models.** Square-zero and split product algebras have a genuine coordinatewise gap, but they do not embed faithfully as scalar subalgebras of $\mathbb{R}$.
- **Structured computation.** Nilpotent truncations show apparent compression; every tested honest real or field lift restores near-injectivity. This is hypothesis generation only.
- **Adversarial audit.** Same-set use, cross-pairs, denominators, embeddings, and rate comparisons were checked independently. No cubic claim survives with $b < 2a$; the companion theorem does.

The central exact identity emerging from the algebraic branch is more useful than any of the cosmetic rewrites. Put

$$H_3(u) = u^3 - 3u, \qquad A_u = A \cap (A-u) = \{y \in A : y+u \in A\}.$$

Then

$$\boxed{P_3(A,A) = \bigcup_{u \in A-A} (H_3(u) - 6A_u).} \tag{C.1}$$

Every slice has cardinality $|A_u|$, and

$$\sum_{u \in A-A} |A_u| = |A|^2. \tag{C.2}$$

Thus a power saving must come from polynomially large overlap between different difference-slices. A collision between the u- and u'-slices is exactly

$$6(y - y') = (u - u')(u^2 + uu' + u'^2 - 3), \tag{C.3}$$

where $y \in A_u$ and $y' \in A_{u'}$. This is the precise global correlation missing from the mechanisms tested below.

D. Best new constructive mechanism

No valid scalar-real construction for P_3 was found. The best conceptual gain is visible in finite tensor algebras, and its failure under real realization identifies the missing information exactly.

Let

$$\mathcal{D}_d = \mathbb{Q} \oplus V, \qquad V^2 = 0,$$

where V has basis $\varepsilon_1, \dots, \varepsilon_d$, and take

$$\mathcal{A}_d = \left\{ \sum_{i=1}^{d} a_i \varepsilon_i : a_i \in \{0,1\} \right\}.$$

Then $x^3 = 0$ for $x \in V$, so

$$P_3(x,y) = -3(x+y).$$

Consequently

$$|\mathcal{A}_d| = 2^d, \qquad |P_3(\mathcal{A}_d, \mathcal{A}_d)| = 3^d,$$

and the formal exponent is

$$c_{\text{formal}} = 2 - \log_2 3 > 0.$$

Likewise, in the split algebra $\mathbb{R}^d$, the alphabet $\Sigma = \{0,1,2,3\}$ has

$$|P_3(\Sigma,\Sigma)| = 14 < 16,$$

which tensorizes to $14^d < 16^d$.

These are not constructions in the problem. A subring of $\mathbb{R}$ is an integral domain, so it has neither a nonzero nilpotent nor a nontrivial idempotent. Thus neither tensor algebra admits a faithful algebra embedding into scalar reals.

The obstruction is sharper than that slogan. In a transcendental lift, with $x = T^k f$ and $y = T^k g$,

$$P_3(x, y) = -3T^k(f + g) + T^{3k}(f - g)^3. \tag{D.1}$$

When a truncation discards all degrees at least d and $3k \geq d$, the second term disappears and creates the formal gain. Before truncation, the linear and cubic degree supports are disjoint. Equality of two outputs recovers $f+g$ and $(f-g)^3$ separately, hence recovers $f-g$ over a real domain and therefore recovers the ordered pair. The deleted cubic tail contains exactly the entropy that the nilpotent model appeared to save.

Specially tailored, highly 3-ramified real number fields remain logically open; they were not ruled out by this argument.

E. Full quantitative analysis of the cubic mechanisms

E.1 Polynomial-coordinate and fixed-norm audit

With

$$u = x - y, \qquad z = P_3(x, y),$$

the map $(x, y) \mapsto (u, z)$ is a polynomial automorphism with constant Jacobian -6 and inverse

$$x = \frac{u^3 + 3u - z}{6}, \qquad y = \frac{u^3 - 3u - z}{6}. \tag{E.1}$$

Hence the labelled sets

$$\{(x - y, P_3(x, y)) : x, y \in A\}$$

and

$$\{(x + y, P_3(x, y)) : x, y \in A\}$$

both have exactly $|A|^2$ elements. The second assertion follows because u is the unique real cube root of $P_3 + 3(x + y)$.

There are exact identities

$$H_3(w + w^{-1}) = w^3 + w^{-3},$$

$$P_3 = \det \begin{pmatrix} u & 0 & -3v \\ 1 & u & 0 \\ 0 & 1 & u \end{pmatrix}, \qquad v = x + y,$$

and

$$P_3 = \mathrm{Res}_T(T - u, T^3 - 3v).$$

The last is a norm in the algebra $L[T]/(T^3 - 3v)$, but this algebra varies with the pair through v. Retaining v restores all $|A|^2$ labels.

There is no replacement by a polynomial norm from one fixed nontrivial field extension. Precisely, let K/L be a nontrivial finite separable extension in characteristic zero. There are no nonconstant $\psi \in K[x, y]$ and $F \in L[t]$ with

$$P_3 = F(N_{K/L}(\psi)).$$

Over an algebraic closure, $P_3 = u^3 - 3v$ is irreducible because it is linear in v with unit leading coefficient. If $\deg F > 1$, splitting F factors the right side. If F is linear, then a translate $P_3 - b$ would equal a product of the at least two nonconstant conjugates of ψ, again contradicting irreducibility. Degree comparison also shows that a direct norm identity could never use growing extension degree: $[K : L] \deg \psi \leq 3$.

This proposition is deliberately scoped to polynomial ψ in a fixed field extension. It does not exclude rational parametrizations, a family of algebras varying with (x, y), or a genuinely new combination of norm and trace.

Two natural identity-driven sets are maximally bad:

- If $T > 1$ is transcendental and $A_n = \{T^i + T^{-i} : 1 \leq i \leq n\}$, the two largest Laurent exponents of each off-diagonal output recover the ordered pair, so $|P_3(A_n, A_n)| = n^2$.
- If $A_n = \{a + \delta i : 0 \leq i < n\}$ and $\delta^2 \notin \mathbb{Q}$, equality of two outputs gives $\delta^2(d^3 - d'^3) = 3(s - s')$ for integer differences and sums. Irrationality recovers d, s, hence the ordered pair. Again the image has size n^2.

A natural growing Galois orbit also fails exactly. In

$$K = \mathbb{Q}(\sqrt{p_1}, \ldots, \sqrt{p_d}),$$

for distinct positive primes p_i, take

$$A = \left\{ \alpha_\varepsilon = \sum_i \varepsilon_i \sqrt{p_i} : \varepsilon_i \in \{-1, 1\} \right\}.$$

The squarefree radical monomials are a $\mathbb{Q}$-basis, so $|A| = 2^d$. For a pair (ε, η) let $D = \{i : \varepsilon_i \neq \eta_i\}$. In the output, the coefficient of the singleton basis vector $\sqrt{p_i}$ is

$$-6\varepsilon_i \quad (i \notin D),$$

and

$$8\varepsilon_i \left(3 \sum_{j \in D} p_j - 2p_i \right) \quad (i \in D).$$

The first has magnitude 6 and the second has magnitude at least 16. The output therefore recovers D, then every ε_i, and finally η. Hence this orbit also has exactly $|A|^2$ outputs.

E.2 Unit-distance and number-field candidate

The supplied unit-distance architecture produces norm-one elements in a CM extension. In the notation of that construction, let L be totally real of degree f, let $K = L(i)$, choose a fixed collection of s split primes, put

$$Q = \prod_{j=1}^{s} q_j, \qquad D = Q^2,$$

and let H be the fixed exponential class-number base. The supplied class-group fiber gives a set $U \subset D^{-1}\mathcal{O}_K$ of norm-one elements satisfying

$$|U| \geq \left(\frac{2^s}{H}\right)^f.$$

Define the genuine same-set candidate

$$A = \left\{ \frac{u + u^{-1}}{2} : u \in U \right\} \subset L \subset \mathbb{R}. \tag{E.2}$$

The trace map has fibers of size at most two: from

$$u + u^{-1} = w + w^{-1}$$

one gets $(u - w)(uw - 1) = 0$. Therefore

$$|A| \geq \frac{1}{2}\left(\frac{2^s}{H}\right)^f. \tag{E.3}$$

Every real conjugate of an element of A lies in $[-1, 1]$. Since

$$P_3(x, y) = H_3(x - y) - 6y$$

and $|H_3(t)| \leq 2$ for $|t| \leq 2$, every conjugate of every output has absolute value at most 8. Also, with $E = 2D$,

$$A \subset E^{-1}\mathcal{O}_L, \qquad P_3(A, A) \subset E^{-3}\mathcal{O}_L.$$

If two elements of $E^{-3}\mathcal{O}_L$ are distinct, their difference is at least E^{-3} in some real embedding: after multiplication by E^3 its nonzero algebraic norm is a nonzero integer. Sup-norm packing now gives the full Cartesian image bound

$$|P_3(A, A)| \leq (16E^3 + 1)^f = (1 + 128D^3)^f. \tag{E.4}$$

Put

$$\alpha = \frac{2^s}{H}, \qquad \beta = 1 + 128D^3.$$

This candidate is an arbitrarily large family only when the supplied tower and the chosen fixed split-prime set give $\alpha > 1$. If that inequality is unavailable, (E.2) should be read only as a finite ledger candidate. The rate obstruction below holds even in the favorable case $\alpha > 1$.

Since $H \geq 1$ and $Q \geq 2^s$,

$$\alpha^2 \leq D, \qquad \frac{\beta}{\alpha^2} > 128D^2. \tag{E.5}$$

Thus this rigorously controlled full-image container loses exponentially. This is a failure of the available counting ledger, not a lower bound on the actual image of the trace set.

The product-window version fails for a structural reason. Let $E_0 \subset \mathbb{R}$ be bounded measurable, let $V = |E_0|$, and put $F = P_3(E_0, E_0)$. Then

$$\boxed{|F + [-1/2, 1/2]| \geq V^2.} \tag{E.6}$$

For $V \leq 3 + \sqrt{10}$, the diagonal inclusion $-6E_0 \subset F$ and the one-dimensional Brunn-Minkowski inequality give

$$|F + [-1/2, 1/2]| \geq 6V + 1 \geq V^2.$$

For $V \geq 4$, choose $y \in E_0$ near its essential infimum and restrict x to $E_0 \cap (y + V/2, \infty)$. This subset has measure arbitrarily close to $V/2$, while

$$\frac{\partial}{\partial x} P_3(x, y) = 3(x - y)^2 - 3 \geq \frac{3V^2}{4} - 3.$$

The one-dimensional change-of-variables formula yields

$$|F| \geq \frac{3V^3}{8} - \frac{3V}{2} \geq V^2.$$

The ranges overlap. Consequently Blichfeldt averaging in a totally real tower, followed only by ambient output packing, cannot give $b < 2a$. For the fractional lattice $D^{-1}\mathcal{O}_L$, the exact local ledger is

$$D^3 \left| F + \left[-\frac{1}{2D^3}, \frac{1}{2D^3} \right] \right| \geq D^2 V^2, \tag{E.7}$$

already at the squared input-volume base before root-discriminant loss. Here is the needed proof with the smaller thickening. If $V \geq 4$, the monotone-slice argument above gives $|F| \geq V^2$, so the left side of (E.7) is at least $D^3 V^2 \geq D^2 V^2$. If $0 < V < 4$, the diagonal and Brunn-Minkowski give

$$\left| F + \left[-\frac{1}{2D^3}, \frac{1}{2D^3} \right] \right| \geq 6V + D^{-3}.$$

After multiplication by D^3, this is $6D^3 V + 1 > D^2 V^2$ for every integer $D \geq 1$.

The substantial standard results in this paragraph are used in these precise forms:

1. One-dimensional Brunn-Minkowski: $|X + Y| \geq |X| + |Y|$ for nonempty measurable $X, Y \subset \mathbb{R}$.
2. Monotone change of variables: if a C^1 map is injective on an interval, then $|g(S)| = \int_S |g'|$ for measurable S in that interval.
3. Blichfeldt averaging: a bounded measurable $W \subset \mathbb{R}^f$ has a translate containing at least $\mathrm{vol}(W)/\operatorname{covol}(\Lambda)$ points of a full lattice Λ.
4. Algebraic norm separation: a nonzero algebraic integer has nonzero integral norm, so at least one conjugate has absolute value at least 1.

E.3 Direct insertion into the supplied sum-product construction

For the frozen construction $A = GP$, with its fixed parameter $0 < \varepsilon \le 1$, write $x = g_1p_1$ and $y = g_2p_2$. Then

$$\begin{aligned} P_3(x,y) = g_1^3p_1^3 - 3g_1^2g_2p_1^2p_2 + 3g_1g_2^2p_1p_2^2 - g_2^3p_2^3 \\ - 3g_1p_1 - 3g_2p_2. \end{aligned} \tag{E.8}$$

The associated unit tuple and carrier tuple each retain their original ordered pair through their last coordinates. Hence their exact joint parameter count is

$$|G|^2|P|^2 = |A|^2. \tag{E.9}$$

This contrasts with multiplication xy, where the two unit variables merge into $g_1g_2 \in GG$. In the cubic, two independent unit-character directions remain.

The archimedean container is worse. If the frozen construction has

$$|\sigma(a)| \le 2Xe^Y,$$

then

$$|\sigma(P_3(x,y))| \le 65X^3e^{3Y},$$

and packing gives the output base $131X^3e^{3Y}$. Against the asymptotic per-degree upper input base $(5Y{+}1)(2\varepsilon X{+}1)$ (the frozen finite bound also has a harmless exterior factor 10), one has for $X, Y \ge 2$

$$\frac{131X^3e^{3Y}}{\big((5Y+1)(2\varepsilon X+1)\big)^2} > 139. \tag{E.10}$$

Thus the box bound is strictly worse than quadratic. The natural exact code above is merely quadratic; it does not rule out additional collisions among its labels. To improve it, one would need a new exponential overlap theorem for the slices in (C.1), equivalently for

$$3(s-s') = (d-d')(d^2+dd'+d'^2).$$

The frozen separate estimates for $A+A$ and AA do not control this four-input correlation.

E.4 High-dimensional digit theorems

Let $\Sigma \subset \mathbb{R}$ have size $h \ge 2$, let θ be transcendental over $\mathbb{Q}(\Sigma)$, and set

$$A_d = \left\{ \sum_{i=0}^{d-1} a_i\theta^i : a_i \in \Sigma \right\}.$$

Then

$$\boxed{|P_3(A_d, A_d)| \ge h^{2d} - h^{d+\lceil d/2 \rceil}.} \tag{E.11}$$

To prove this, represent two pairs by polynomials a, b and a', b', put $u = a-b$, $v = a+b$, and similarly u', v'. Equality gives

$$(u-u')(u^2+uu'+u'^2)=3(v-v'). \tag{E.12}$$

The right side has degree at most $d-1$. If $u \neq u'$ and $D=\max(\deg u, \deg u')$, then the positive-definite leading form X^2+XY+Y^2 shows that the second factor has degree $2D$. Therefore equality is impossible if either difference has degree at least $\lceil d/2 \rceil$. All such ordered pairs have unique outputs. The excluded pairs agree in their top $\lfloor d/2 \rfloor$ digits, and their number is exactly $h^{d+\lceil d/2 \rceil}$.

There is also an exact good-reduction statement. Let $\mathcal{O}$ be a torsion-free $\mathbb{Z}$-order for which $\mathcal{O}/3\mathcal{O}$ is reduced, and let $A \subset \mathcal{O}$ reduce injectively modulo $3\mathcal{O}$. Then P_3 is injective on A^2. Indeed, equality of outputs gives $u^3=u'^3$ modulo 3. Frobenius is injective in a reduced characteristic-three ring, so $u \equiv u' \pmod 3$. Expanding $u=u'+3w$ then gives $u^3-u'^3 \in 9\mathcal{O}$, whence $v \equiv v' \pmod 3$ and finally $x \equiv x'$, $y \equiv y' \pmod 3$. Injectivity of the chosen residue section finishes the proof.

Ordinary carries do not help for full digit alphabets. For $A_n=\{0,1,\dots,n-1\}$, difference-slices with $|u| \geq 3\sqrt{n}$ occupy disjoint intervals, giving

$$|P_3(A_n,A_n)| \geq n^2-O(n^{3/2}). \tag{E.13}$$

E.5 Offline structured computation

Computation was used only to reject or refine candidates. It was not used as proof.

- Formal polynomial digits with $\Sigma=\{0,1\}$ gave exactly 4^d outputs for $d \leq 8$.
- Three-symbol formal alphabets with distinct residues modulo 3 gave exactly 9^d outputs for $d \leq 5$.
- The formal collision alphabet $\Sigma=\{0,3,6\}$ gave $7, 75, 711, 6507, 58887, 530955$ outputs for $d=1,\dots,6$; its ratio to 9^d rose throughout the tested range.
- The truncated nilpotent binary model gave $4, 10, 34, 112, 380, 1292, 5488, 20112$ outputs for $d=1,\dots,8$.
- Cyclic wraps $T^d=\pm 1$ with binary digits had full or near-full pair image for $d \leq 9$.
- Quadratic algebraic recurrences had an observed image exponent rising toward 2 for $d \leq 9$.
- Integer intervals were near-quadratic, consistently with the theorem (E.13).

The first two rows are explained by the exact mod-3 theorem. The attractive truncated row is explained by the deleted tail in (D.1). The collision alphabet shows why a one-coordinate defect does not automatically tensor under scalar polynomial digits: cross-coordinate convolution restores the missing information.

F. Independent audit

The final claims were checked against the following failure modes.

- **Distinct real inputs:** transcendental evaluations and field embeddings used above are injective; the companion affine map is a real bijection.
- **Same set in both variables:** every positive theorem uses one common A'_n in both slots. Rich bipartite subgraphs were not promoted to same-set image bounds.
- **Full Cartesian square:** (E.4) and the companion theorem count every pair. Prescribed fiber incidences were rejected because unmatched cross-pairs remain.
- **Energy versus image:** no collision-energy lower bound is used as an image upper bound. Equation (C.1) is a direct union identity.
- **Denominators:** in (E.2), $A \subset (2D)^{-1}\mathcal{O}_L$ and outputs lie in $(2D)^{-3}\mathcal{O}_L$; the cubic denominator is fully charged.
- **Embeddings:** a field embedding is injective. Norm separation is applied only after clearing the exact denominator.
- **Class number and root discriminant:** they enter the supplied norm-one theorem through one fixed exponential base H; no subexponential saving is claimed.

- **Constants:** $Q, D, H, X, Y, \varepsilon$ are fixed while field degree grows. All failed ledgers fail at the fixed per-degree base.
- **Nilpotents and split coordinates:** their formal gain is excluded from the scalar-real theorem because no faithful algebra embedding into $\mathbb{R}$ exists.
- **Special-form scope:** the non-special claims concern polynomial $F, g, h,$ with both inner polynomials nonconstant. Constant Φ is excluded from the companion non-special theorem.
- **Computational evidence:** every computational statement is labeled as evidence only and is unnecessary for the proved companion theorem.

No contradiction with the frozen local results was found. The present no-go statements concern specific global architectures; they are not presented as a disproof of CPSI.

G. Proved companion theorem

Theorem G.1: postcomposition and common-affine inheritance

Let $S \in \mathbb{R}[x, y]$. Suppose there are $c_0 > 0$ and finite sets $A_n \subset \mathbb{R}$, with $|A_n| \to \infty$, such that

$$|S(A_n, A_n)| \leq |A_n|^{2-c_0}.$$

For fixed $\Phi \in \mathbb{R}[t]$, $a \neq 0$, and $b \in \mathbb{R}$, put

$$Q(x, y) = \Phi(S(ax + b, ay + b))$$

and

$$A'_n = \frac{A_n - b}{a} = \left\{ \frac{t - b}{a} : t \in A_n \right\}.$$

Then $|A'_n| = |A_n|$ and

$$Q(A'_n, A'_n) = \Phi(S(A_n, A_n)).$$

Because a function cannot increase the cardinality of a finite image,

$$|Q(A'_n, A'_n)| \leq |S(A_n, A_n)| \leq |A'_n|^{2-c_0}. \tag{G.1}$$

All constants are unchanged and independent of n.

Apply the theorem with $S = R$. Since

$$R(ax + b, ay + b) = a^2(x - y)^2 + 5a(x + y) + 10b, \tag{G.2}$$

the resulting explicit family is

$$\boxed{Q_{\Phi,a,b}(x, y) = \Phi\big(a^2(x - y)^2 + 5a(x + y) + 10b\big).} \tag{G.3}$$

For nonconstant Φ, its partial-derivative ratio is

$$\frac{(Q_{\Phi,a,b})_x}{(Q_{\Phi,a,b})_y} = \frac{2a(x - y) + 5}{5 - 2a(x - y)}. \tag{G.4}$$

This is a nonconstant rational function of $x - y$, so Lemma B.1 proves that every polynomial in (G.3) is outside both special forms. Its degree is $2 \deg \Phi$.

For the requested concrete polynomial, take $a = 1$, $b = 0$, and $\Phi(t) = t^2 + t$. Then

$$\boxed{P_4(x, y) = \big((x-y)^2 + 5(x+y)\big)^2 + (x-y)^2 + 5(x+y).}$$

It has degree 4, is non-special, and the original frozen witness sets for R satisfy

$$|P_4(A_n, A_n)| \le |R(A_n, A_n)| \le |A_n|^{2-c_0}.$$

Thus $c = c_0$ is an explicit inherited exponent in terms of the already-proved seed theorem.

H. Verdict

3. CPSI remains open, but an explicit new polynomial family with a proved power saving has been established.

The proved family is (G.3), and every nonconstant member is non-special in the polynomial sense. The cubic work did not merely stop at unsuccessful experiments: it isolated the exact translated-slice union (C.1), ruled out a fixed field-norm amplification, proved near- or exact injectivity for broad digit models, and showed quantitatively why the direct number-field and $A = GP$ containers do not cross the entropy threshold.

I. Exactly one next mathematical target

Fix one explicit witness sequence $A_n^{(R)}$ from the proved theorem for R. Prove or refute the existence of a constant $c > 0$ such that, for arbitrarily large n,

$$\boxed{\left| \bigcup_{u \in A_n^{(R)} - A_n^{(R)}} \Big(H_3(u) - 6\big(A_n^{(R)} \cap (A_n^{(R)} - u)\big)\Big) \right| \le |A_n^{(R)}|^{2-c}.}$$

This is CPSI restricted to the already successful quadratic-seed sets, and it asks for exactly the cross-slice overlap missing from the current sum-product and number-field estimates.